



# [Document title]

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# Contents

|                                                                   |    |
|-------------------------------------------------------------------|----|
| 1. Introduction .....                                             | 4  |
| 2. Industrial Task Overview .....                                 | 5  |
| 2.1 Description of the industrial task .....                      | 5  |
| 2.2 Process sequence narrative .....                              | 5  |
| 2.3 Functional objective .....                                    | 6  |
| 3. Key Elements to Be Considered in PLC Program Preparation ..... | 7  |
| 3.1 Functional requirements table .....                           | 7  |
| 3.2 Signal types .....                                            | 7  |
| 3.3 Number systems .....                                          | 8  |
| 3.4 Key logic functions .....                                     | 8  |
| 3.5 Associated program elements .....                             | 17 |
| 4. PLC Input and Output Allocation .....                          | 18 |
| 4.1 I/O allocation list .....                                     | 18 |
| 4.2 Input device explanation .....                                | 18 |
| 4.3 Output device explanation .....                               | 19 |
| 5. Communication Connections .....                                | 20 |
| 5.1 Overview of communication connections .....                   | 20 |
| 5.2 Communication methods table .....                             | 20 |
| 5.3 Correct use of communication connections .....                | 21 |
| 6. PLC Program Design .....                                       | 22 |
| 6.1 Program structure .....                                       | 22 |
| 6.2 Logic diagram or screenshots .....                            | 22 |
| 6.3 Rationale for the design .....                                | 22 |
| 7. Testing Strategy .....                                         | 24 |
| 7.1 Test plan .....                                               | 24 |
| 7.2 Testing method .....                                          | 24 |
| 7.3 Debugging method .....                                        | 25 |
| 8. Test Results and Fault Corrections .....                       | 26 |
| 8.1 Test results record .....                                     | 26 |
| 8.2 Fault correction examples .....                               | 26 |
| 8.3 Evidence of improvement .....                                 | 26 |
| 9. Fully Functional PLC Design and Performance Analysis .....     | 27 |
| 9.1 Final program summary .....                                   | 27 |
| 9.2 Performance analysis .....                                    | 27 |
| 9.3 Design strengths and limitations .....                        | 27 |

|                                         |    |
|-----------------------------------------|----|
| 9.4 Scope for further development ..... | 28 |
| 10. Conclusion .....                    | 29 |
| Bibliography.....                       | 30 |

**No table of figures entries found.**

# 1. Introduction

Write 3 to 5 lines explaining what this task is trying to achieve.

You should explain:

- what industrial task the PLC is being designed for
- what functions the PLC must perform
- that the workbook will show design, communication, testing and debugging
- that the final outcome is a fully functional PLC design with performance analysis

Suggested sentence starters:

- This lab workbook presents the design of a PLC program for an automated handling cell by identifying the required functions, allocating inputs and outputs, selecting communication methods, and recording testing and debugging activities.
- The aim of the task is to create a fully functional PLC design for a given industrial task and evaluate its performance in operation.
- The workbook documents the full engineering process from requirements capture to testing, fault correction and final performance review.

Insert a system overview diagram showing:

- conveyor or process station
- sensors
- actuators
- PLC
- HMI if used
- communication links
- operator controls
- emergency stop or fault signals if relevant

## 2. Industrial Task Overview

### 2.1 Description of the industrial task

Write a short paragraph explaining the application.

You should include:

- what the system does
- what product or object is being handled
- what the PLC is controlling
- what the start and end conditions of the sequence are

Suggested sentence starters:

- The industrial task used for this PLC design is...
- The PLC is required to control...
- The process begins when...
- The process ends when...

### 2.2 Process sequence narrative

Write the complete control sequence as numbered steps. Aim for 8 to 12 steps.

Example structure:

1. Operator presses the start push button.
2. The PLC checks the system is in a safe and ready condition.
3. The conveyor or feed mechanism moves the workpiece into position.
4. A sensor detects the part at the required location.
5. The PLC energises the actuator or output device.
6. A timer controls the action duration.
7. A confirmation sensor or state bit verifies completion.
8. The output is de-energised.
9. The system returns to ready state.
10. If a fault condition occurs, the process stops and a fault signal is generated.

Insert a flowchart or sequence diagram showing:

- idle
- start
- automatic operation

- completion
- fault state
- reset state

## 2.3 Functional objective

Write a brief subsection summarising the required behaviour of the program.

You should state:

- what the PLC must do
- what conditions it must check
- what outputs it must control
- how faults are handled

## 3. Key Elements to Be Considered in PLC Program Preparation

This section is essential for P4.

### 3.1 Functional requirements table

Create a table with rows such as:

- system start condition
- stop condition
- reset condition
- sensor detection condition
- actuator output condition
- timer requirement
- counter requirement
- latch or memory requirement
- fault condition
- communication requirement
- operator indication requirement

Recommended table headings:

- required function
- description
- related input or output
- logic element needed
- notes

Insert a completed requirements table.

### 3.2 Signal types

Explain the signal types used in the system.

You should include:

- digital inputs
- digital outputs
- analogue signals if used
- internal memory bits

- timer values
- counter values

Suggested sentence starters:

- The system uses digital input signals for...
- Digital outputs are used to control...
- Internal memory bits are required for...
- If analogue signals are used, they represent...

### 3.3 Number systems

<https://edge.autonomylogic.com/docs/openplc-editor/iec-concepts/variables-datatypes>

Include a short section on number systems where relevant.

You may refer to:

- binary
- decimal
- hexadecimal
- addressed I/O formats
- bit-level representation in PLC memory

This section does not need to be long, but it should show awareness of how signals and addressing are represented.

Suggested sentence starters:

- PLC inputs and outputs are commonly addressed in...
- Binary representation is useful for...
- Hexadecimal notation may be used for...

### 3.4 Key logic functions

Explain the main logic functions required in the design.

Include:

## AND

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour

OR

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour

## XOR

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour

## NOT

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour

## Latching

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour

## Counting

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour

## Timing

For each one, explain what role it plays in the control sequence. Include truth table and logic symbols.

Suggested sentence starters:

- An AND function is required when...
- An OR function is used to...
- A latch is needed to...
- Interlocking prevents...

Include the Simulation diagram

Explain briefly the logic behaviour



### 3.5 Associated program elements

Explain the supporting program elements that must be considered.

Include:

- timers
- counters
- latches
- set and reset functions
- internal relays or markers
- sequencing logic

Write short paragraphs explaining why each one is needed.

## 4. PLC Input and Output Allocation

### 4.1 I/O allocation list

Produce a full input and output allocation table.

<https://edge.autonomylogic.com/docs/openplc-editor/hardware-configuration/pin-mapping>

Recommended columns:

- address
- tag name
- device description
- type
- normal state
- function in sequence

Example rows:

- I0.0 Start PB
- I0.1 Stop PB
- I0.2 Reset PB
- I0.3 Part Sensor
- Q0.0 Conveyor Motor
- Q0.1 Pneumatic Actuator
- Q0.2 Buzzer
- Q0.3 Indicator Lamp

Insert the completed I/O allocation table.

### 4.2 Input device explanation

Write a short explanation of each major input device.

You should include:

- push buttons
- switches
- sensors

- limit devices
- fault signals

### 4.3 Output device explanation

Write a short explanation of each major output device.

You should include:

- motors
- solenoids
- relays
- buzzers
- lamps
- drives if used

## 5. Communication Connections

This section is essential for P5.

<https://edge.autonomylogic.com/docs/openplc-editor/communication/modbus/server>

<https://edge.autonomylogic.com/docs/openplc-editor/communication/modbus/client>

<https://edge.autonomylogic.com/docs/openplc-editor/communication/modbus/addressing>

### 5.1 Overview of communication connections

Explain how communication connections are correctly used with the PLC.

You should include the communication links present in your design, for example:

- PLC to HMI
- PLC to remote I/O
- PLC to VSD
- PLC to another controller
- PLC to engineering workstation

Suggested sentence starters:

- Communication connections are used in this system to...
- The PLC communicates with...
- The selected connection method is suitable because...

### 5.2 Communication methods table

Create a table with headings such as:

- device connected
- communication method
- purpose
- data exchanged
- reason for selection

Examples:

- HMI, Ethernet, operator control and status display
- Remote I/O, fieldbus, distributed sensor data

- VSD, industrial Ethernet or serial, speed command and status

### 5.3 Correct use of communication connections

Write short paragraphs explaining:

- why the connection type is appropriate
- what configuration is required
- how addressing or network setup is managed
- what problems could occur if communication is incorrect

Suggested sentence starters:

- This communication method is appropriate because...
- Correct configuration requires...
- If the communication link is incorrectly configured...

Insert a simple network diagram showing PLC and connected devices.

## 6. PLC Program Design

### 6.1 Program structure

Explain how the PLC program is structured.

You may divide it into sections such as:

- start and stop logic
- sequencing logic
- timer and counter logic
- fault logic
- reset logic
- output control
- communication handling

Suggested sentence starters:

- The PLC program is divided into functional sections to improve...
- The first section manages...
- The next section controls...

### 6.2 Logic diagram or screenshots

Insert ladder logic, function block diagrams or structured text screenshots.

You should include:

- start and stop network
- sensor logic
- timer or counter logic
- output control
- fault or interlock logic

Under each figure, write 2 to 4 lines explaining what the logic does.

### 6.3 Rationale for the design

Write 2 to 4 paragraphs explaining why the program was designed in this way.

You should refer to:

- simplicity
- reliability
- clarity

- ease of maintenance
- suitability for the industrial task

Suggested sentence starters:

- The design approach was chosen because...
- A timer was used instead of...
- Interlocking was included to...
- The program structure supports troubleshooting by...

## 7. Testing Strategy

This section supports M2.

### 7.1 Test plan

Create a test plan before running the program.

Recommended columns:

- test number
- test condition
- expected result
- actual result
- pass or fail
- comments

Include tests such as:

- power-up and initial state
- start sequence
- sensor response
- actuator operation
- stop operation
- reset operation
- fault condition
- communication status
- timer behaviour
- counter behaviour

Insert the completed test plan table.

### 7.2 Testing method

Explain how you tested the PLC hardware and software.

You should refer to:

- simulation
- online monitoring
- forcing inputs or outputs where permitted
- checking status bits

- step-by-step sequence testing
- observing output response
- checking communication diagnostics

Suggested sentence starters:

- Testing was carried out systematically by...
- The first stage of testing involved...
- Online monitoring was useful because...
- Fault simulation helped to verify...

## 7.3 Debugging method

Explain the debugging methods used.

Include:

- tracing logic conditions
- checking incorrect addresses
- verifying timer presets
- checking counter values
- confirming interlock conditions
- checking communication status
- correcting wiring assumptions in software models if relevant

Suggested sentence starters:

- Debugging was carried out by...
- A fault was identified when...
- The cause of the issue was traced to...
- The correction improved program operation by...

## 8. Test Results and Fault Corrections

### 8.1 Test results record

Summarise the outcome of testing.

You can use a table with:

- test performed
- result obtained
- issue identified
- action taken
- final outcome

### 8.2 Fault correction examples

Describe at least two faults, errors or issues found during testing.

Examples:

- incorrect input address
- timer value too short
- output latched incorrectly
- communication tag mismatch
- sensor logic reversed
- sequence failed to reset correctly

For each one, explain:

- what went wrong
- how it was identified
- what correction was made
- what the final result was

### 8.3 Evidence of improvement

Show how testing and debugging improved the final program.

This is important for Merit.

Suggested sentence starters:

- After debugging, the system operated more reliably because...
- The correction to the timer logic ensured...
- Communication stability improved when...

## 9. Fully Functional PLC Design and Performance Analysis

This section is essential for D2.

### 9.1 Final program summary

Summarise the final version of the PLC design.

Include:

- main functions achieved
- final sequence operation
- communication performance
- handling of normal and fault conditions

### 9.2 Performance analysis

Write 3 to 5 short paragraphs analysing the performance of the PLC design.

You should consider:

- sequence accuracy
- response time
- output timing
- reliability
- communication behaviour
- fault response
- ease of use and maintainability

Suggested sentence starters:

- The final PLC design is fully functional because...
- The sequence operated correctly under...
- Performance was limited slightly by...
- Communication behaviour was acceptable because...
- The design would be improved further by...

### 9.3 Design strengths and limitations

Write a balanced analysis.

Include:

- strengths of the design

- limitations or remaining risks
- how the limitations could be addressed in future

## 9.4 Scope for further development

Briefly explain what could be added if the system were expanded.

Examples:

- HMI visualisation
- alarm history
- remote monitoring
- additional sensors
- analogue control
- recipe selection
- SCADA integration

## 10. Conclusion

Write one concise conclusion summarising:

- what was designed
- how communication was used
- how testing and debugging were integrated
- whether the PLC design met the industrial task requirements

Suggested sentence starter:

- In conclusion, the PLC program was designed by considering the required functional elements, communication connections, and systematic testing and debugging methods, resulting in a working control solution for the given industrial task.

## Bibliography

**There are no sources in the current document.**